Cairo University

Faculty of Engineering

Electronics and Electrical Communications Engineering

Assignment 1

ELC 4028 – Artificial Neural Networks

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PART 1: Regression Problems

Problem 1: Insurance Company

The numbers of insured persons y with an insurance company for the years 1987 to 1996 are given. We let x represent the number of years since 1987 ($x = 0$ for 1987, $x = 9$ for 1996).

1.1 Dataset

x (years since 1987)	Year	y (insured persons)
0	1987	12,400
1	1988	10,900
2	1989	10,000
3	**1990**	**1,050 (OUTLIER)**
4	1991	9,500
5	1992	8,900
6	1993	8,000
7	1994	7,800
8	1995	7,600
9	1996	7,200

Table 1: Insurance dataset (1987–1996)

** Value 1050 at $x = 3$ (year 1990) is the outlier — likely a data entry error or anomaly.*

1.2 Case A: WITH Outlier

1. Scatter Plot & Outlier Identification A scatter plot of y (number of insured persons) versus x (years since 1987) reveals a generally decreasing trend from $\approx$ 12,400 down to $\approx$ 7,200. However, one data point stands out sharply:

- $x = 3$ (year 1990): $y = 1050$

This value is dramatically lower than the surrounding data, which are all within approximately 7,200 to 12,400. The sharp drop breaks the otherwise smooth downward trend. This observation is therefore identified as a clear outlier, most likely due to a data entry error (e.g. 10,500 instead of 1,050).

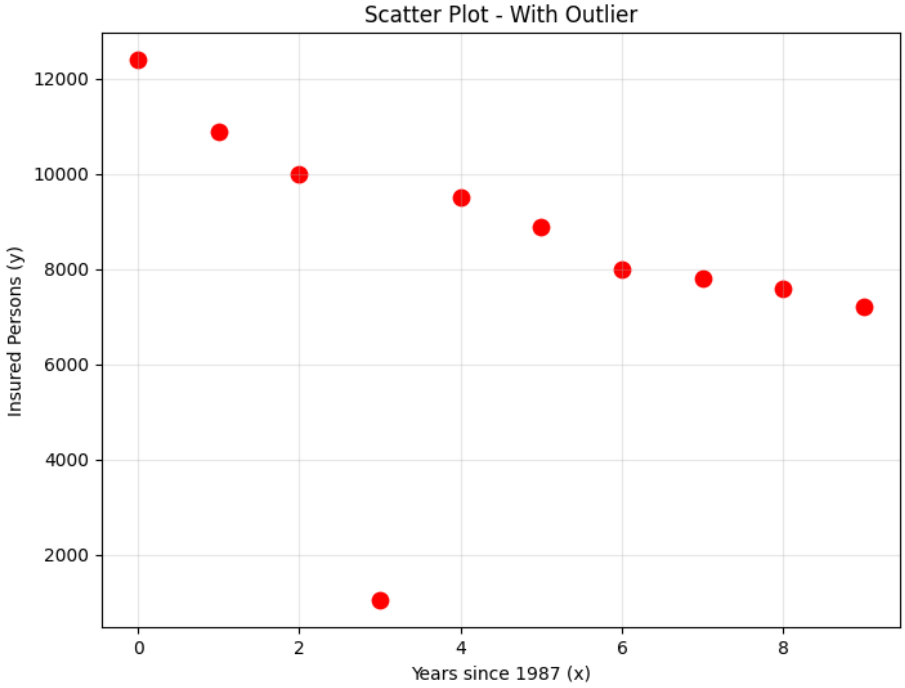


Figure 1: Scatter plot of original dataset identifying the outlier at $x = 3$ ($y = 1050$)

2. Regression Models (WITH Outlier) We apply both OLS and matrix-form methods to compute linear model parameters, then use the matrix form for quadratic and cubic models.

2.1 Linear Model Hand Analysis (OLS Method)

The equation for the linear model is:

$$\hat{y}_i = \beta_0 + \beta_1 x_i \tag{1}$$

The key formulas:

$$\hat{\beta}_1 = \frac{\displaystyle\sum_{i=1}^{n} x_i y_i - \frac{\left(\sum x_i\right)\left(\sum y_i\right)}{n}}{\displaystyle\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum x_i\right)^2}{n}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1\,\bar{x} \tag{2}$$

Table 2: Data table for hand computation (WITH outlier)

Year	x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1987	0	12,400	0	153,760,000	0
1988	1	10,900	1	118,810,000	10,900
1989	2	10,000	4	100,000,000	20,000
1990	3	1,050	9	1,102,500	3,150
1991	4	9,500	16	90,250,000	38,000
1992	5	8,900	25	79,210,000	44,500
1993	6	8,000	36	64,000,000	48,000
1994	7	7,800	49	60,840,000	54,600
1995	8	7,600	64	57,760,000	60,800
1996	9	7,200	81	51,840,000	64,800
Σ	**45**	**83,350**	**285**	**777,572,500**	**344,750**

$n = 10, \quad \bar{y} = 83350/10 = 8335, \quad \bar{x} = 45/10 = 4.5$

$$\hat{\beta}_1 = \frac{344{,}750 - \dfrac{45 \times 83{,}350}{10}}{285 - \dfrac{45^2}{10}} = -367.57$$

$$\hat{\beta}_0 = 8335 + 367.57 \times 4.5 = 9989.09$$

Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T Y$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 9989.09 \\ -367.576 \end{pmatrix}$$

Linear equation:

$$\hat{y}_i = 9989.09 - 367.576\, x \tag{3}$$

R² Verification Table

$$\bar{y} = \frac{\sum y_i}{n} = \frac{83350}{10} = 8335$$

Table 3: R^2 verification for linear model (WITH outlier)

Year	x_i	y_i	$\hat{y}_i = \beta_0 + \beta_1 x_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	9,989.1	5,812,483	16,524,225
1988	1	10,900	9,621.5	1,634,524	6,579,225
1989	2	10,000	9,253.9	556,606	2,772,225
1990	3	1,050	8,886.4	61,408,600	53,071,225
1991	4	9,500	8,518.8	962,777	1,357,225
1992	5	8,900	8,151.2	560,683	319,225
1993	6	8,000	7,783.6	46,813	112,225
1994	7	7,800	7,416.1	147,410	286,225
1995	8	7,600	7,048.5	304,169	540,225
1996	9	7,200	6,680.9	269,455	1,288,225
Σ	45	83,350	83,350	**71,703,515**	**82,850,250**

$$R^2_{\text{linear}} = \frac{\text{SS}_{yy} - \text{SSE}}{\text{SS}_{yy}} = \frac{82{,}850{,}250 - 71{,}703{,}515}{82{,}850{,}250} = 0.1345$$

Output Graph

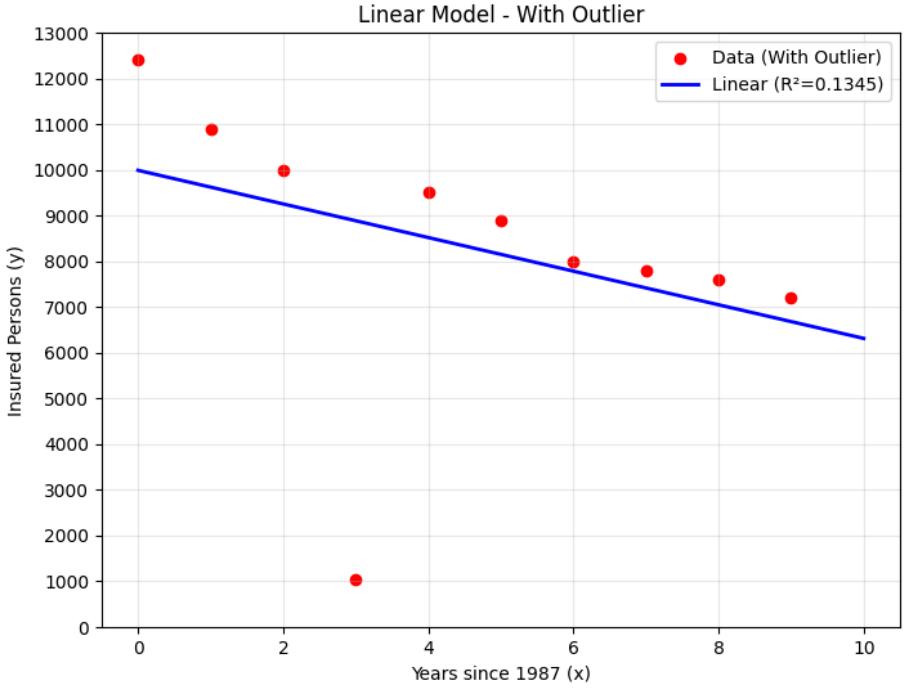


Figure 2: Linear Model of Original Data with Outlier

Simulation Results

```
Linear Model:
  Equation: y = 9989.09 + (-367.58)*x^1
  R^2: 0.1345
```

Figure 3: Simulation Results of Linear Model of Original Data with Outlier

2.2 Quadratic Model The equation for the quadratic model:

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 \tag{4}$$

Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 & 0^2 \\ 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \\ 1 & 6 & 6^2 \\ 1 & 7 & 7^2 \\ 1 & 8 & 8^2 \\ 1 & 9 & 9^2 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 11627.73 \\ -1596.55 \\ 136.55 \end{pmatrix}$$

Quadratic equation:

$$\hat{y}_i = 11627.73 - 1596.55\,x + 136.55\,x^2 \tag{5}$$

R² Verification Table

Table 4: R^2 verification for quadratic model (WITH outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	11,627.7	596,405	16,524,225
1988	1	10,900	10,167.7	536,223	6,579,225
1989	2	10,000	8,980.8	1,038,701	2,772,225
1990	3	1,050	8,067.0	49,238,925	53,071,225
1991	4	9,500	7,426.4	4,299,969	1,357,225
1992	5	8,900	7,058.8	3,390,064	319,225
1993	6	8,000	6,964.3	1,072,639	112,225
1994	7	7,800	7,143.0	431,710	286,225
1995	8	7,600	7,594.7	28	540,225
1996	9	7,200	8,319.5	1,253,377	1,288,225
Σ	45	83,350	83,350	**61,858,042**	**82,850,250**

$$R^2_{\text{quadratic}} = \frac{82{,}850{,}250 - 61{,}858{,}042}{82{,}850{,}250} = 0.2534$$

Output Graph

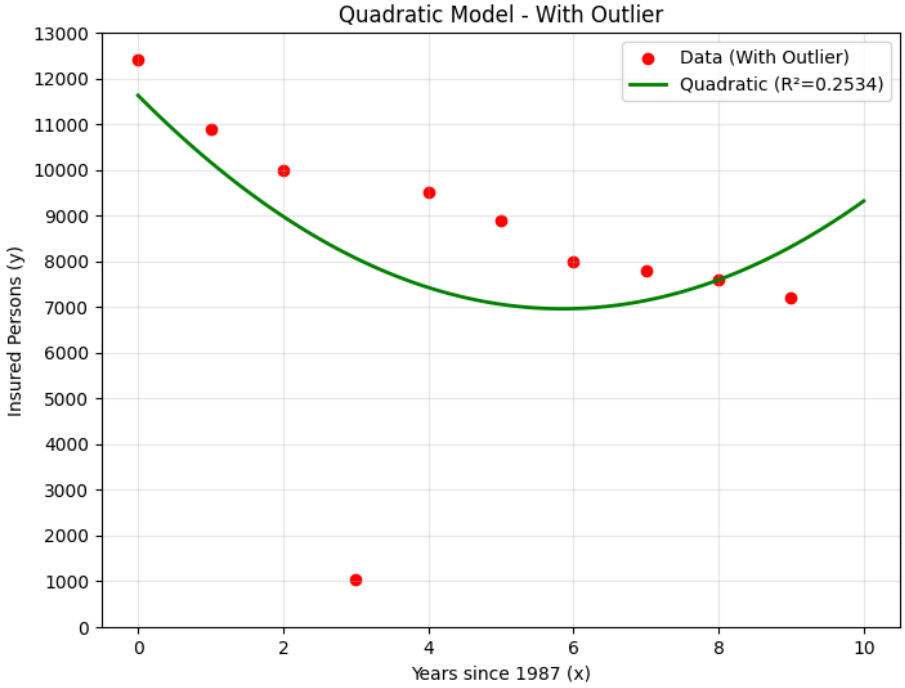


Figure 4: Quadratic Model of Original Data with Outlier

Simulation Results

```
Quadratic Model:
  Equation: y = 11627.73 + (-1596.55)*x^1 + (136.55)*x^2
  R^2: 0.2534
```

Figure 5: Simulation Results of Quadratic Model of Original Data with Outlier

2.3 Cubic Model The equation for the cubic model:

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 \tag{6}$$

Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T Y$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 & 0^2 & 0^3 \\ 1 & 1 & 1^2 & 1^3 \\ 1 & 2 & 2^2 & 2^3 \\ 1 & 3 & 3^2 & 3^3 \\ 1 & 4 & 4^2 & 4^3 \\ 1 & 5 & 5^2 & 5^3 \\ 1 & 6 & 6^2 & 6^3 \\ 1 & 7 & 7^2 & 7^3 \\ 1 & 8 & 8^2 & 8^3 \\ 1 & 9 & 9^2 & 9^3 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} 13113.15 \\ -4313.93 \\ 932.31 \\ -58.95 \end{pmatrix}$$

Cubic equation:

$$\hat{y}_i = 13113.15 - 4313.93\, x_i + 932.31\, x_i^2 - 58.95\, x_i^3 \tag{7}$$

R² Verification Table

Table 5: R^2 verification for cubic model (WITH outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	13,113.1	508,578	16,524,225
1988	1	10,900	9,672.6	1,506,542	6,579,225
1989	2	10,000	7,743.0	5,094,123	2,772,225
1990	3	1,050	6,970.7	35,054,266	53,071,225
1991	4	9,500	7,002.0	6,240,214	1,357,225
1992	5	8,900	7,483.2	2,007,341	319,225
1993	6	8,000	8,060.7	3,684	112,225
1994	7	7,800	8,380.8	337,334	286,225
1995	8	7,600	8,089.8	239,940	540,225
1996	9	7,200	6,834.1	133,864	1,288,225
Σ	45	83,350	83,350	**51,125,885**	**82,850,250**

$$R^2_{\text{cubic}} = \frac{82{,}850{,}250 - 51{,}125{,}885}{82{,}850{,}250} = 0.3829$$

Output Graph

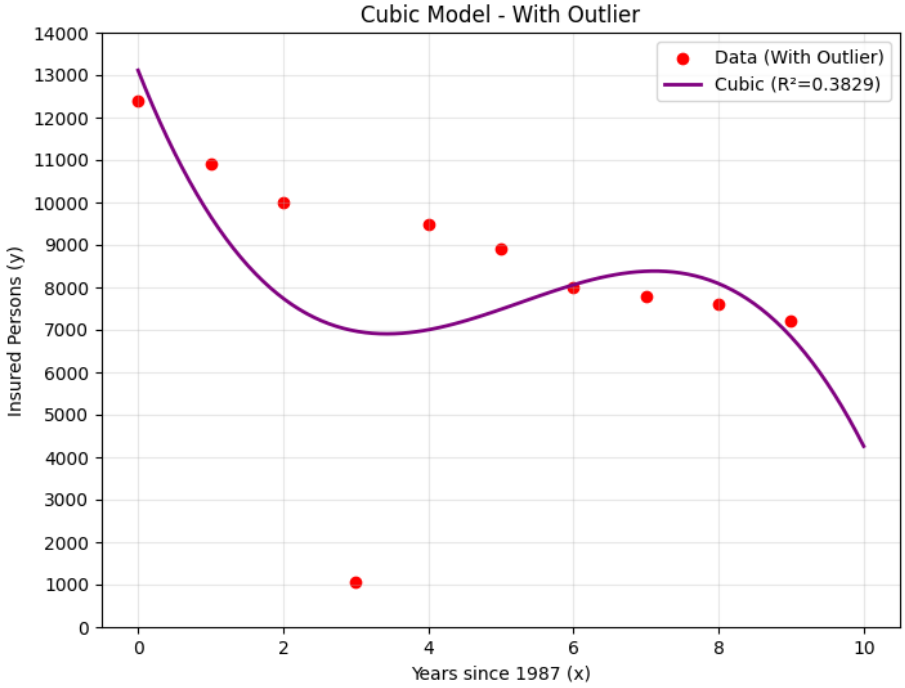


Figure 6: Cubic Model of Original Data with Outlier

Simulation Results

```
Cubic Model:
  Equation: y = 13113.15 + (-4313.93)*x^1 + (932.31)*x^2 + (-58.95)*x^3
  R^2: 0.3829
```

Figure 7: Simulation Results of Cubic Model of Original Data with Outlier

Table 6: Model comparison – WITH outlier

Model	**Equation**	R^2
Linear	$\hat{y}_i = 9989.09 - 367.576\,x_i$	0.1345
Quadratic	$\hat{y}_i = 11627.73 - 1596.55\,x_i + 136.55\,x_i^2$	0.2534
Cubic	$\hat{y}_i = 13113.15 - 4313.93\,x_i + 932.31\,x_i^2 - 58.95\,x_i^3$	0.3829

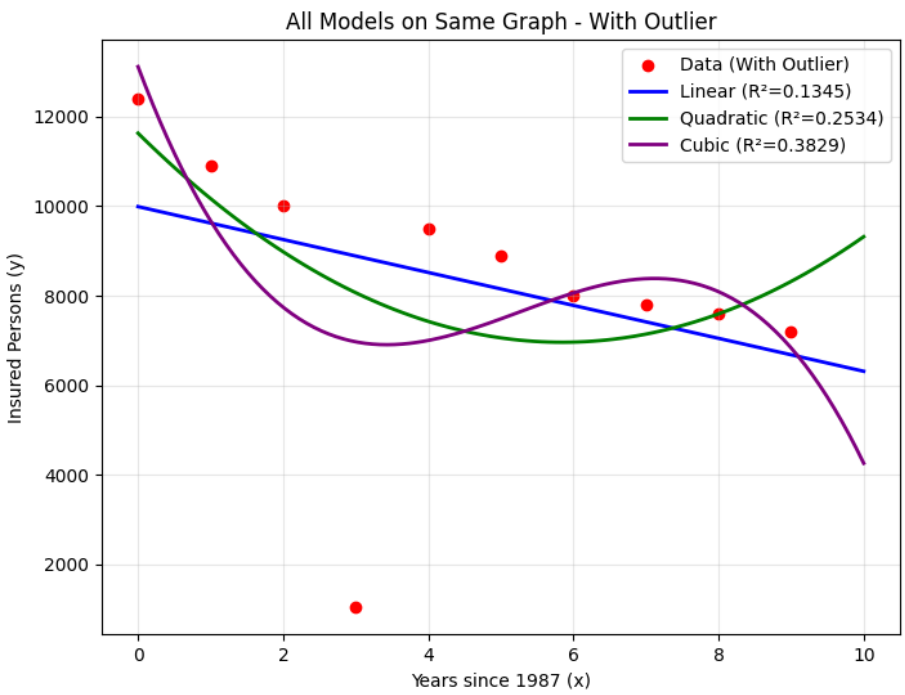


Figure 8: Comparison between three Models of Original Data with Outlier

3. Comparison Between Three Models (WITH Outlier) **Comment:** All R^2 values are low (< 0.50) due to the heavy influence of the outlier at $x = 3$ ($y = 1050$). The cubic model yields the highest R^2.

1.3 Case B: WITHOUT Outlier

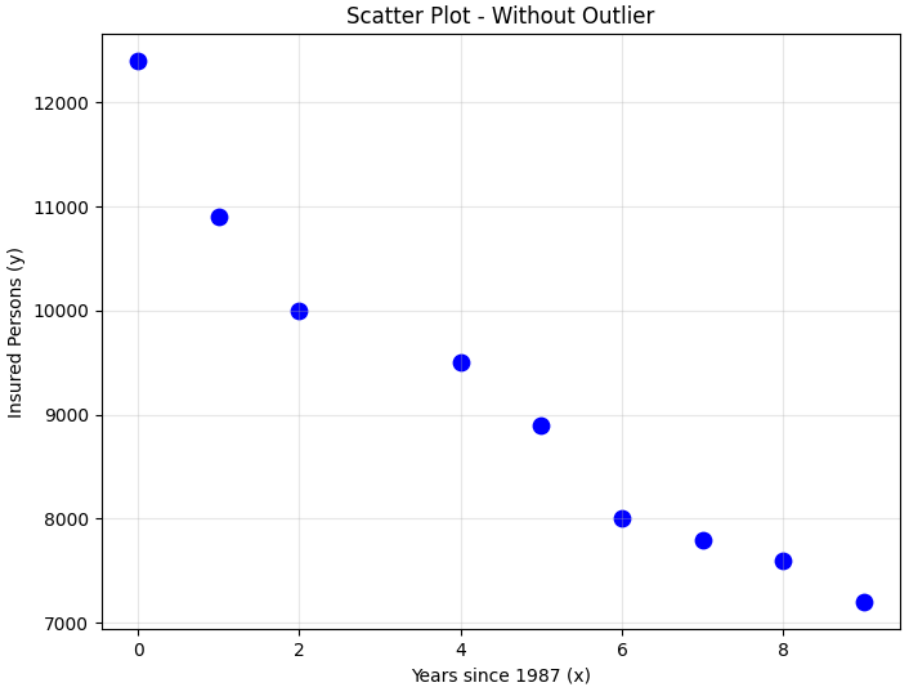


Figure 9: Scatter Plot of Cleaned Data without Outlier

1. Outlier Removal and Scatter Plot of Cleaned Data

2. Regression Models (WITHOUT Outlier) After removing the outlier at $x = 3$ ($y = 1{,}050$), the cleaned dataset contains 9 data points. We now repeat the regression analysis using the same methods.

2.1 Linear Model Hand Analysis (OLS Method)

$$\hat{y}_i = \beta_0 + \beta_1 x_i$$

Table 7: Data table for hand computation (WITHOUT outlier)

Year	x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1987	0	12,400	0	153,760,000	0
1988	1	10,900	1	118,810,000	10,900
1989	2	10,000	4	100,000,000	20,000
1991	4	9,500	16	90,250,000	38,000
1992	5	8,900	25	79,210,000	44,500
1993	6	8,000	36	64,000,000	48,000
1994	7	7,800	49	60,840,000	54,600
1995	8	7,600	64	57,760,000	60,800
1996	9	7,200	81	51,840,000	64,800
Σ	**42**	**82,300**	**276**	**776,470,000**	**341,600**

$n = 9, \quad \bar{y} = 82300/9 = 9{,}144.44, \quad \bar{x} = 42/9 = 4.667$

$$\hat{\beta}_1 = \frac{341{,}600 - \dfrac{42 \times 82{,}300}{9}}{276 - \dfrac{42^2}{9}} = -530.83$$

$$\hat{\beta}_0 = 9{,}144.44 + 530.83 \times 4.667 = 11{,}621.67$$

Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T Y$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 11{,}621.67 \\ -530.83 \end{pmatrix}$$

Linear equation:

$$\hat{y}_i = 11{,}621.67 - 530.83\, x$$

R^2 Verification Table

$$\bar{y} = \frac{82300}{9} = 9{,}144.44$$

Table 8: R^2 verification for linear model (WITHOUT outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	11,621.7	605,803	10,598,642
1988	1	10,900	11,090.8	36,417	3,081,975
1989	2	10,000	10,560.0	313,600	731,975
1991	4	9,500	9,498.3	3	126,420
1992	5	8,900	8,967.5	4,556	59,753
1993	6	8,000	8,436.7	190,678	1,309,753
1994	7	7,800	7,905.8	11,201	1,807,531
1995	8	7,600	7,375.0	50,625	2,385,309
1996	9	7,200	6,844.2	126,617	3,780,864
Σ	42	82,300	82,300.0	**1,339,500**	**23,882,222**

$$R^2_{\text{linear}} = \frac{23{,}882{,}222 - 1{,}339{,}500}{23{,}882{,}222} = 0.9439$$

Output Graph

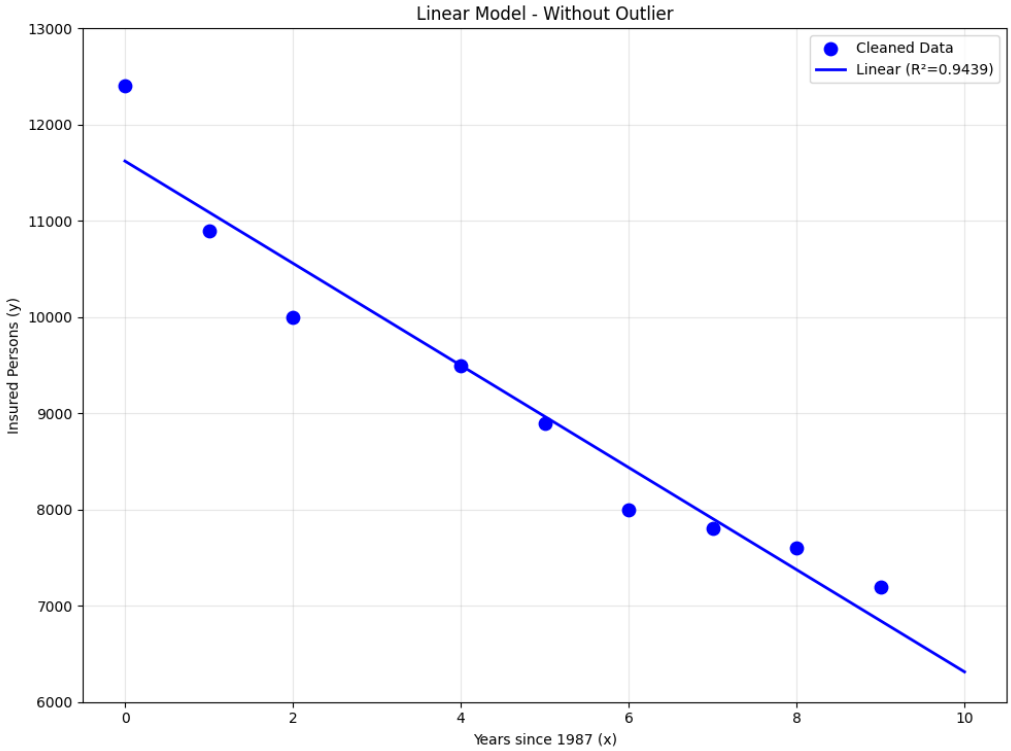


Figure 10: Linear Model of Cleaned Data without Outlier

Simulation Results

```
Linear Model:
  Equation: y = 11621.67 + (-530.83)*x^1
  R^2: 0.9439
```

Figure 11: Simulation Results of Linear Model of Cleaned Data without Outlier

2.2 Quadratic Model

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$$

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T Y, \quad \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 12{,}024.17 \\ -863.13 \\ 37.44 \end{pmatrix}$$

Quadratic equation:

$$\hat{y}_i = 12{,}024.17 - 863.13\,x + 37.44\,x^2$$

$\mathbf{R^2}$ Verification Table

$$\bar{y} = 82{,}300/9 = 9{,}144.44$$

Table 9: R^2 verification for quadratic model (WITHOUT outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	12,024.2	141,248	10,598,642
1988	1	10,900	11,198.5	89,090	3,081,975
1989	2	10,000	10,447.7	200,411	731,975
1991	4	9,500	9,170.7	108,429	126,420
1992	5	8,900	8,644.6	65,249	59,753
1993	6	8,000	8,193.3	37,362	1,309,753
1994	7	7,800	7,816.9	286	1,807,531
1995	8	7,600	7,515.4	7,156	2,385,309
1996	9	7,200	7,288.8	7,884	3,780,864
Σ	42	82,300	82,300.0	**657,116**	**23,882,222**

$$R^2_{\text{quadratic}} = \frac{23{,}882{,}222 - 657{,}116}{23{,}882{,}222} = 0.9725$$

Output Graph

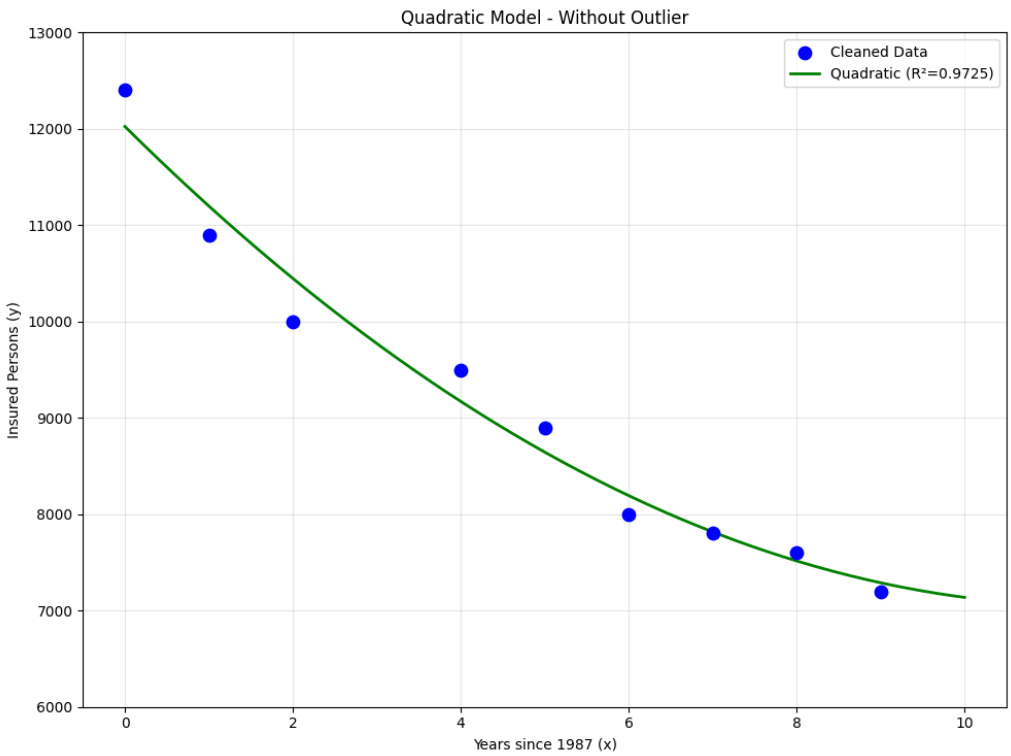


Figure 12: Quadratic Model of Cleaned Data without Outlier

Simulation Results

```
Quadratic Model:
  Equation: y = 12024.17 + (-863.13)*x^1 + (37.44)*x^2
  R^2: 0.9725
```

Figure 13: Simulation Results of Quadratic Model of Cleaned Data without Outlier

2.3 Cubic Model

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} 12{,}204.42 \\ -1{,}221.74 \\ 140.17 \\ -7.46 \end{pmatrix}$$

Cubic equation:

$$\hat{y}_i = 12{,}204.42 - 1{,}221.74\, x_i + 140.17\, x_i^2 - 7.46\, x_i^3$$

R^2 Verification Table

Table 10: R^2 verification for cubic model (WITHOUT outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	12,204.4	38,251	10,598,642
1988	1	10,900	11,115.4	46,392	3,081,975
1989	2	10,000	10,261.9	68,595	731,975
1991	4	9,500	8,666.9	174,339	126,420
1992	5	8,900	8,307.8	54,323	59,753
1993	6	8,000	7,960.3	94,746	1,309,753
1994	7	7,800	7,579.7	25,702	1,807,531
1995	8	7,600	7,121.1	413	2,385,309
1996	9	7,200	6,081.0	6,227	3,780,864
Σ	42	82,300	82,300.0	**508,988**	**23,882,222**

$$R^2_{\text{cubic}} = \frac{23{,}882{,}222 - 508{,}988}{23{,}882{,}222} = 0.9787$$

Output Graph

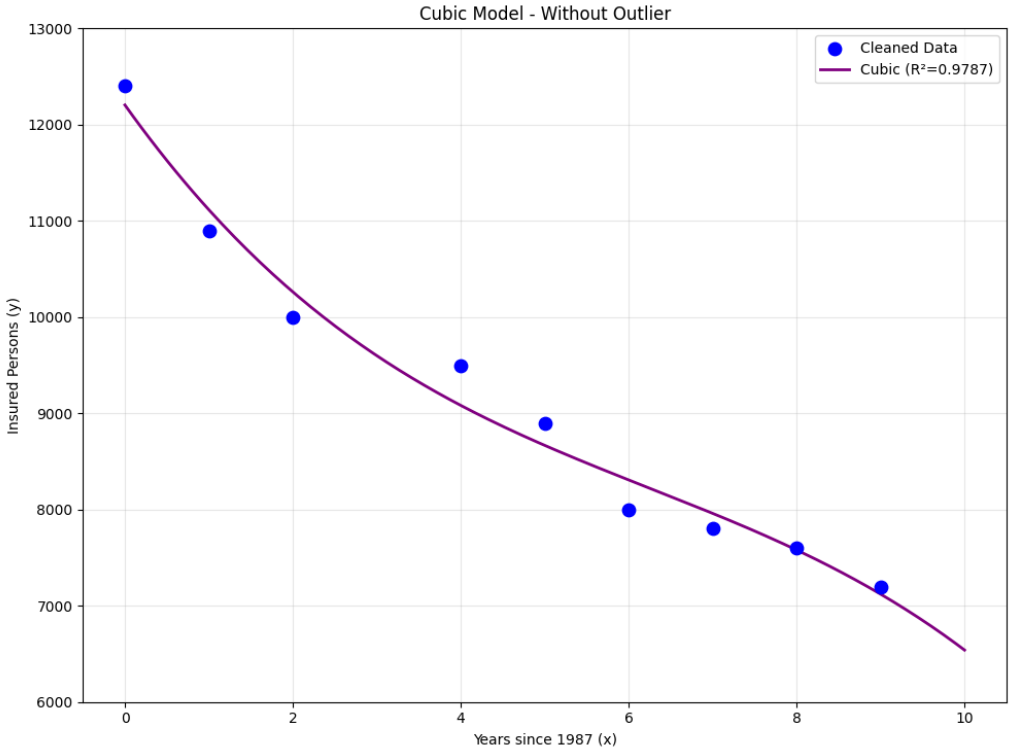


Figure 14: Cubic Model of Cleaned Data without Outlier

Simulation Results

```
Cubic Model:
  Equation: y = 12204.42 + (-1221.74)*x^1 + (140.17)*x^2 + (-7.46)*x^3
  R^2: 0.9787
```

Figure 15: Simulation Results of Cubic Model of Cleaned Data without Outlier

Table 11: Model comparison – WITHOUT outlier (cleaned data)

Model	**Equation**	R^2
Linear	$\hat{y}_i = 11{,}621.67 - 530.83\,x$	0.9439
Quadratic	$\hat{y}_i = 12{,}024.17 - 863.13\,x + 37.44\,x^2$	0.9725
Cubic	$\hat{y}_i = 12{,}204.42 - 1{,}221.74\,x_i + 140.17\,x_i^2 - 7.46\,x_i^3$	0.9787

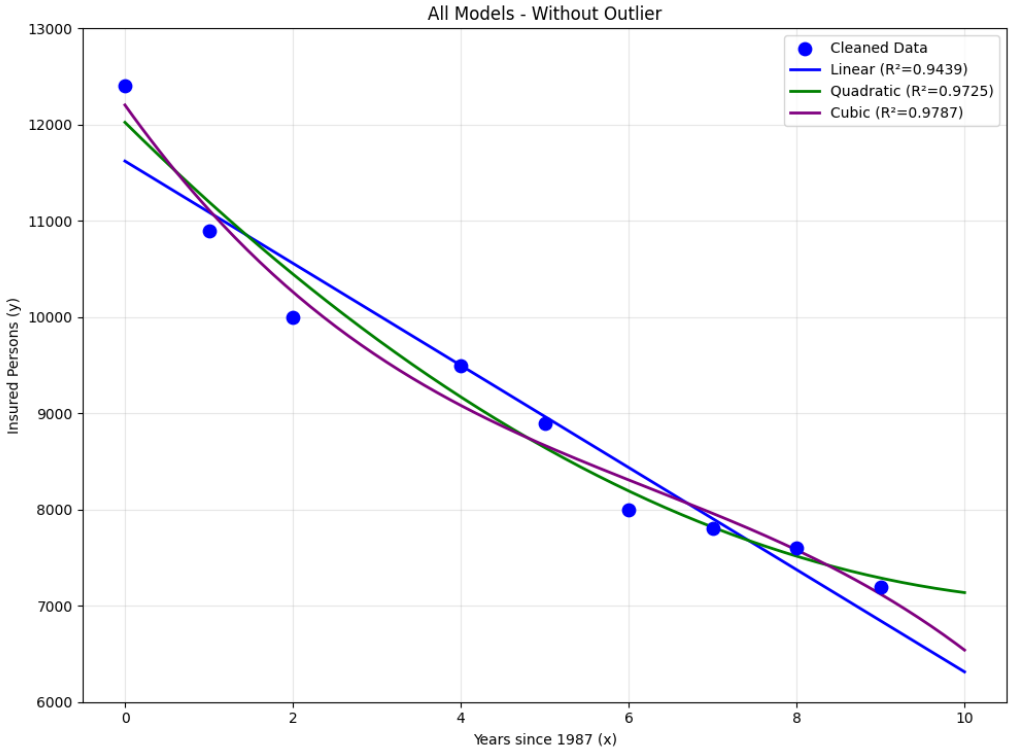


Figure 16: Comparison between three Models of Cleaned Data without Outlier

1.3 Comparison Between Three Models (Cleaned Data) **Comment:** All three R^2 values are very high (> 0.94) after removing the outlier, confirming that it was the primary source of distortion. The linear model alone explains 94.4% of the variance.

1.4 Best-Fit Model Selection

R² Comparison

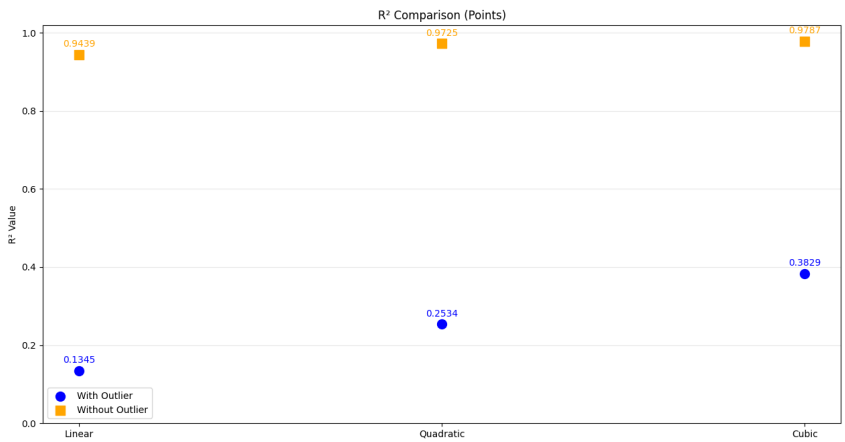


Figure 17: R^2 Comparison between three Models

Table 12: R^2 comparison and model selection

Model	**Params**	R^2 (Original) $\rightarrow$ R^2 (Cleaned)	**Gain (Original)**	**Gain (Cleaned)**
Linear	2	$0.1345 \rightarrow 0.9439$	—	—
Quadratic	3	$0.2534 \rightarrow 0.9725$	$+0.1189$	$+0.0286$
Cubic	4	$0.3829 \rightarrow 0.9787$	$+0.1295$	$+0.0062$

Conclusion (with outlier):

- Linear $\rightarrow$ Quadratic: R^2 gain $= 0.1189$ (significant) $\Rightarrow$ Quadratic is better.
- Quadratic $\rightarrow$ Cubic: R^2 gain $= 0.1295$ (significant) $\Rightarrow$ Cubic is better.

Best-fit model (with outlier): Cubic

$$\hat{y} = 13113.15 - 4313.93\,x + 932.31\,x^2 - 58.95\,x^3 \quad (R^2 = 0.3829)$$

Conclusion (without outlier):

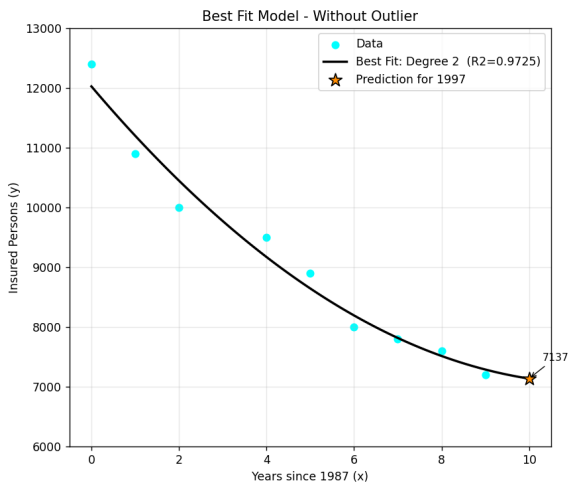
- Linear $\rightarrow$ Quadratic: R^2 gain $= 0.0286$ (significant) $\Rightarrow$ Quadratic is better.
- Quadratic $\rightarrow$ Cubic: R^2 gain $= 0.0062$ $\Rightarrow$ gain is NOT significant enough.

Best-fit model (without outlier): Quadratic

$$\hat{y} = 12{,}024.17 - 863.13\,x + 37.44\,x^2 \quad (R^2 = 0.972)$$

```
>>  Model Selection: Cubic recommended
   dR2 (Linear -> Quadratic) = +0.1188
   dR2 (Quadratic -> Cubic)  = +0.1295
```

(a) Best Fit Model with Outlier



(b) Best Fit Model without Outlier

Figure 18: Best-fit models

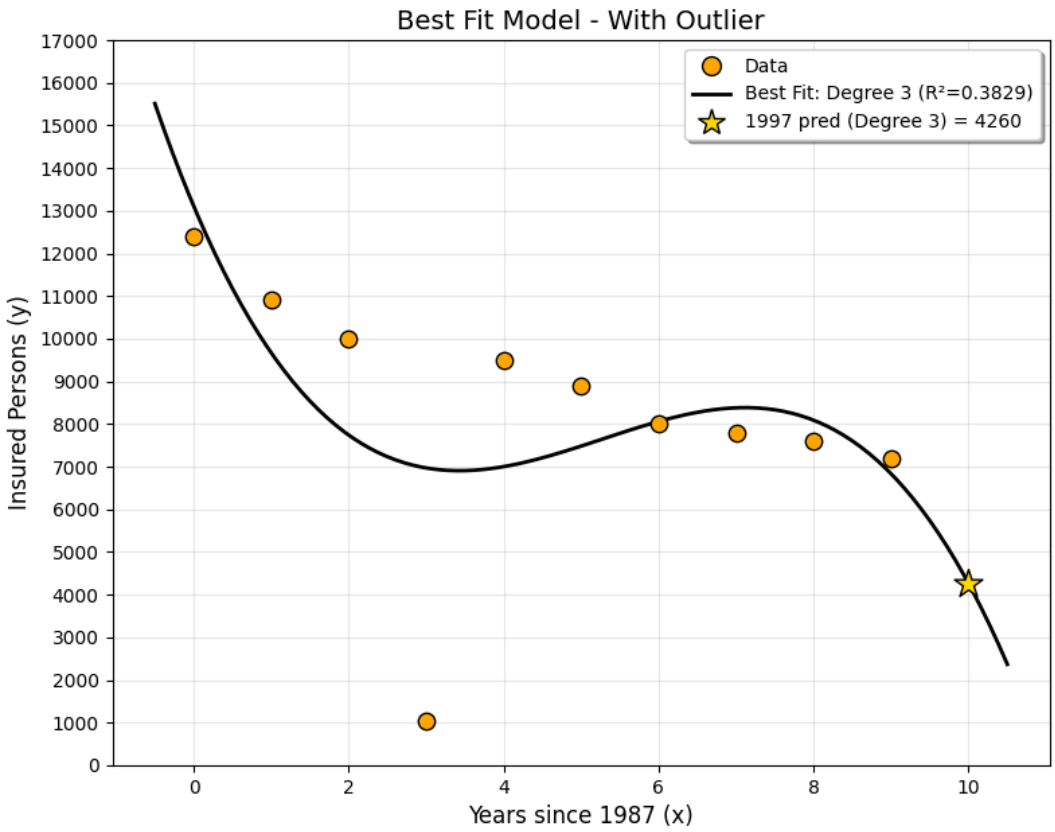


Figure 19: Simulation Results of Best Fit Model with Outlier

```
>>  Model Selection: Quadratic recommended
    dR2 (Linear -> Quadratic) = +0.0286
    dR2 (Quadratic -> Cubic)  = +0.0062
```

Figure 20: Simulation Results of Best Fit Model without Outlier

1.5 Prediction of Insured People

At year 1997 ($x = 10$):

- Using the best-fit model from the original data **with outlier** (Cubic model):

$$\hat{y}_i = 13113.15 - 4313.93\,x_i + 932.31\,x_i^2 - 58.95\,x_i^3$$
$$\hat{y}(10) = 13113.15 - 4313.93(10) + 932.31(10^2) - 58.95(10^3) = 4{,}254.85$$

Insured People $\approx$ 4,254.85 insured persons

- Using the best-fit model from the cleaned data **without outlier** (Quadratic model):

$$\hat{y}_i = 12{,}024.17 - 863.13\,x + 37.44\,x^2$$
$$\hat{y}(10) = 12{,}024.17 - 863.13(10) + 37.44(10^2) = 7{,}136.87$$

Insured People ≈ 7,136.87 insured persons

```
==============================================================
  PREDICTIONS FOR 1997  (year = 1997 , x = 10)
==============================================================
  Original Data with-outlier (Cubic Model) -> 4,260.00 insured persons

  Cleaned Data without-outlier (Quadratic Model) -> 7,137.06 insured persons
```

Figure 21: Simulation Results of Prediction of Insured People at Year 1997

Problem 2: Heating Oil

We develop a regression model to estimate heating oil usage for a single-family home in January, based on two predictors: average temperature (Temp, in °F) and amount of insulation (Insulation, in inches).

2.1 Dataset

Table 13: Heating oil dataset

Oil (gallons)	Temp (°F)	Insulation (in)
270	40	4
362	27	4
162	40	10
45	73	6
91	65	7
233	**65**	**40 ← OUTLIER**
372	10	6
305	9	10
234	24	10
122	65	4
25	66	10
210	41	6
450	22	4
325	40	4
52	60	10

2.2 Case A: WITH Outlier

1. Scatter Plot & Outlier Identification A 3D scatter plot of Oil Usage vs. Temperature and Insulation reveals that one data point is far outside the cluster: (Temp= 65°F,

Insulation= 40 in, Oil= 233 gallons). All other insulation values range from 4 to 10 inches; an insulation thickness of 40 inches is physically unreasonable and is almost certainly a data entry error (e.g., 4 recorded as 40). This point is highlighted in red below.

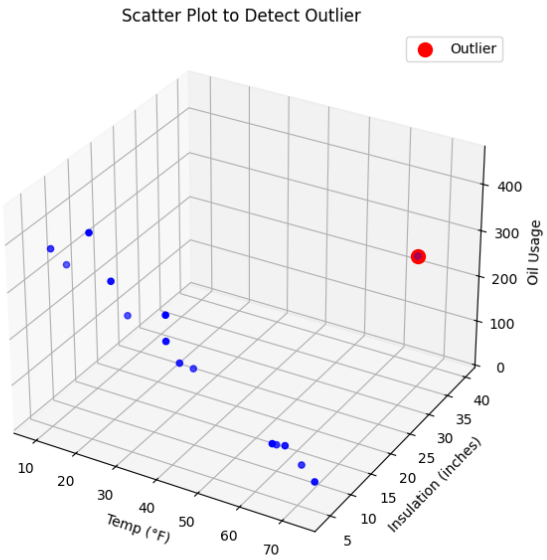


Figure 22: 3D Scatter Plot to Detect Outlier (outlier in red: Insulation= 40)

2. Regression Models

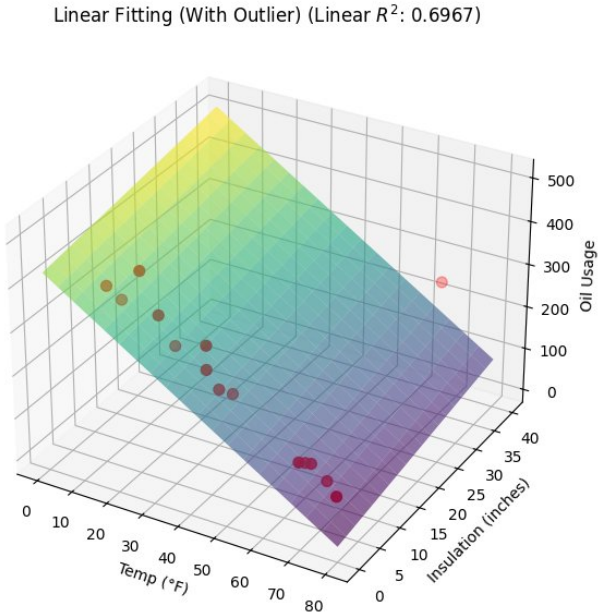


Figure 23: 3D Linear Regression Surface (With Outlier), $R^2 = 0.6967$

2.1 Linear Model The multivariate linear model:

$$\text{Oil} = \beta_0 + \beta_1 T + \beta_2 I \tag{8}$$

where $T = \text{Temp}$, $I = \text{Insulation}$.

$$\text{Oil} = 426.97 - 5.274\,T + 1.966\,I$$

```
Linear model:
  Intercept : 426.967656
  Coef      : Temp=-5.273465,  Insulation=1.966053
  Equation  : y = 426.9677  +  (-5.2735)*Temp  +  (1.9661)*Insulation
  R2        : 0.696686
```

Figure 24: Simulation Results of Linear Regression Surface (With Outlier)

$$\hat{y}_i = 426.97 - 5.27\,x_{1i} + 1.966\,x_{2i}$$

$$R^2 \text{ (Linear, With Outlier)} = 0.6967$$

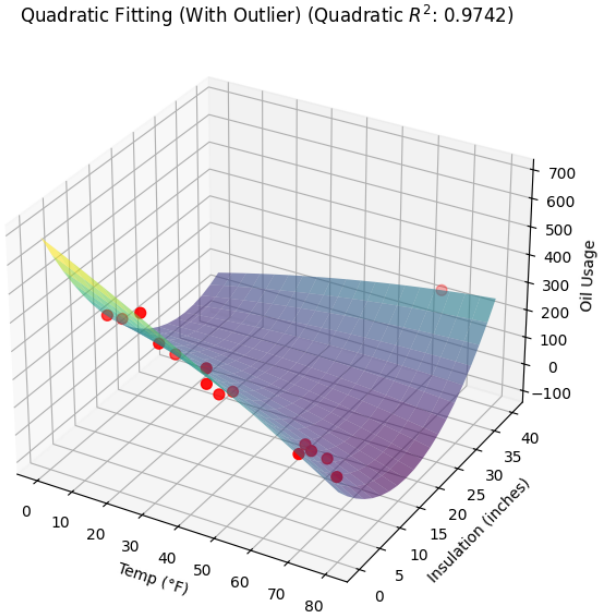


Figure 25: 3D Quadratic Regression Surface (With Outlier), $R^2 = 0.9742$

2.2 Quadratic Model The multivariate quadratic model:

$$\text{Oil} = \beta_0 + \beta_1 T + \beta_2 I + \beta_3 T^2 + \beta_4 TI + \beta_5 I^2 \tag{9}$$

```
Quadratic model:
  Intercept : 680.025958
  Temp                : -6.167784
  Insulation          : -41.716060
  Temp^2              : -0.012475
  Temp Insulation     : 0.239147
  Insulation^2        : 0.658593
  Equation  : y = 680.0260  +  (-6.1678)*Temp  +  (-41.7161)*Insulation  +  (-0.0125)*Temp^2  +  (0.2391)*Temp
  Insulation  +  (0.6586)*Insulation^2
  R2        : 0.974152
```

Figure 26: Coefficient values for quadratic model with outlier

$$\text{Oil} = 680.03 - 6.168\,T - 41.716\,I - 0.01248\,T^2 + 0.2392\,TI + 0.6586\,I^2$$

$$\hat{y}_i = 680 - 6.17\,x_{1i} - 41.72\,x_{2i} - 0.012\,x_{1i}^2 + 0.24\,x_{1i}x_{2i} + 0.66\,x_{2i}^2$$

$$R^2 \text{ (Quadratic, With Outlier)} = 0.9742$$

2.3 Case B: WITHOUT Outlier

1. Scatter Plot (WITHOUT Outlier) The data point (Temp= 65, Insulation= 40, Oil= 233) is removed. All remaining 14 points have insulation values between 4 and 10 inches, which is physically reasonable.

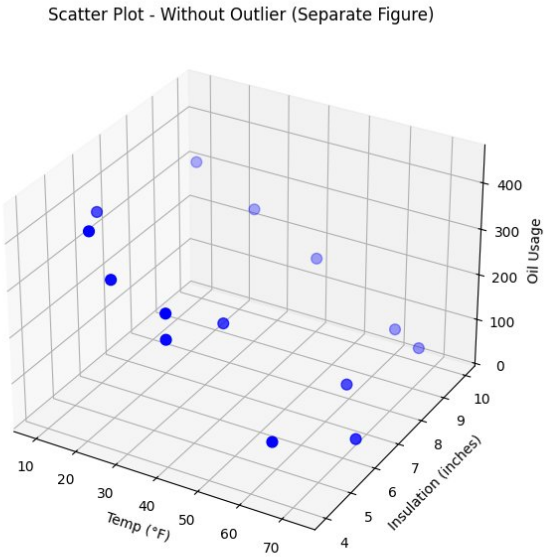


Figure 27: 3D Scatter Plot (Cleaned Data, Outlier Removed)

2. Regression Models (WITHOUT Outlier)

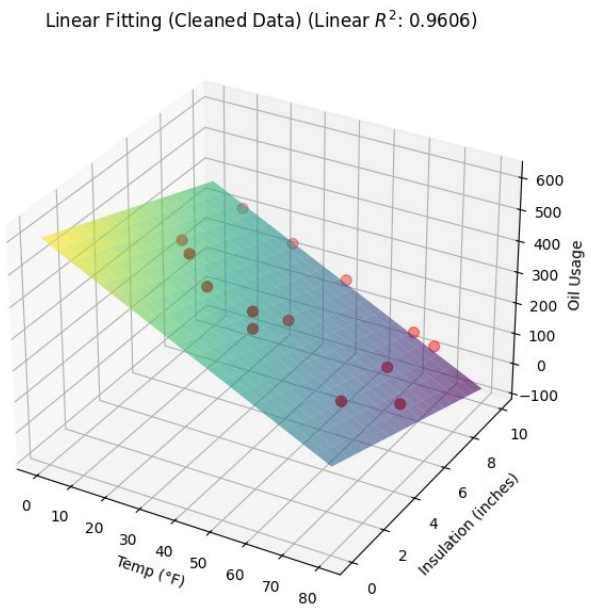


Figure 28: 3D Linear Regression Surface (Cleaned Data), $R^2 = 0.9606$

2.1 Linear Model

$$\text{Oil} = 601.6 - 5.482\,T - 23.232\,I$$

```
Linear model:
  Intercept : 601.601412
  Coef      : Temp=-5.481747,  Insulation=-23.232031
  Equation  : y = 601.6014  +  (-5.4817)*Temp  +  (-23.2320)*Insulation
  R2        : 0.960609
```

Figure 29: Simulation Results of Linear Regression Surface (Cleaned Data)

$$\hat{y}_i = 601.6 - 5.482\,x_{1i} - 23.232\,x_{2i}$$

$$R^2 \text{ (Linear, Without Outlier)} = 0.9606$$

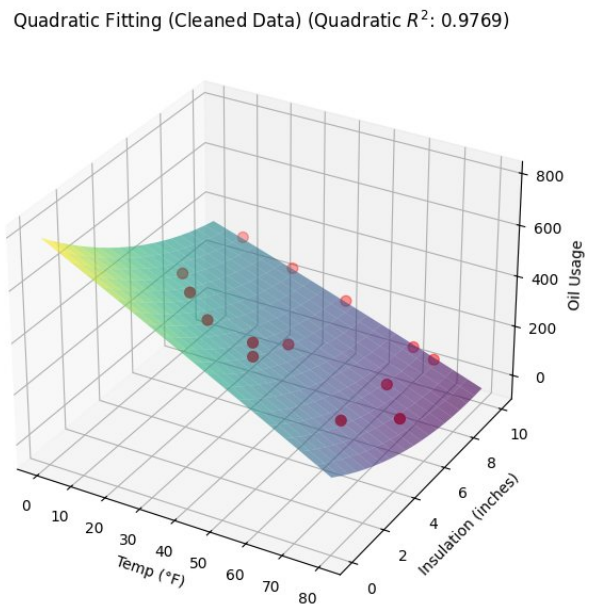


Figure 30: 3D Quadratic Regression Surface (Cleaned Data), $R^2 = 0.9769$

2.2 Quadratic Model

$$\text{Oil} = 785.52 - 6.963\,T - 72.159\,I - 0.00348\,T^2 + 0.2567\,TI + 2.7438\,I^2$$

```
Quadratic model:
  Intercept : 785.516872
  Temp                : -6.962462
  Insulation          : -72.158742
  Temp^2              : -0.003478
  Temp Insulation     : 0.256701
  Insulation^2        : 2.743800
  Equation  : y = 785.5169  +  (-6.9625)*Temp  +  (-72.1587)*Insulation  +  (-0.0035)*Temp^2  +  (0.2567)*Temp
  Insulation  +  (2.7438)*Insulation^2
  R2        : 0.976889
```

Figure 31: Simulation Results of Quadratic Regression Surface (Cleaned Data)

$$\hat{y}_i = 785.52 - 6.963\,x_{1i} - 72.159\,x_{2i} - 0.00348\,x_{1i}^2 + 0.2567\,x_{1i}x_{2i} + 2.7438\,x_{2i}^2$$

$$R^2 \text{ (Quadratic, Without Outlier)} = 0.9769$$

2.4 Comparison of Models and Best-Fit Selection

Table 14: Model comparison for heating oil

Model	R^2 (With Outlier)	R^2 (Without Outlier)	**Gain (With)**	**Gain (Without)**
Linear	0.6967	0.9606	—	—
Quadratic	0.9742	0.9769	+0.2775	+0.0163

Conclusion (Part a – With Outlier):
Linear → Quadratic gain = +0.2775 ⇒ **Best-fit model (with outlier): Quadratic** ($R^2 = 0.9742$).

Conclusion (Part b – Without Outlier):
Linear → Quadratic gain = +0.0163 (not significant) ⇒ **Best-fit model (without outlier): Linear** ($R^2 = 0.9606$).

(a) Best Fit Model of Original Data (Quadratic)

(b) Best Fit Model of Cleaned Data (Linear)

Figure 32: Best-fit surface plots for heating oil

After removing the outlier, the linear model already achieves $R^2 = 0.9606$. The gain from linear to quadratic is only +0.0163, which is NOT significant enough as specified in Note 4 of the assignment.

```
Part a  (With Outlier):
  dR2 (Linear -> Quadratic) = +0.2775  -->  significant gain
  Best-fit model: Quadratic  (R2 = 0.9742)

Part b  (Without Outlier):
  dR2 (Linear -> Quadratic) = +0.0163  -->  not significant
  Best-fit model: Linear     (R2 = 0.9606)
```

Figure 33: Simulation Results of Best Fit Model

2.5 Prediction of Oil

Given: Temp = 15°F, Insulation = 5 inches.

The linear model can be expressed as:

$$\hat{y}_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$$

Table 15: Linear model coefficients and prediction

	With Outlier	**Without Outlier (Best Fit)**
β_0	427	601.6
β_1	-5.273	-5.482
β_2	1.966	-23.23
R^2	0.6966	0.9606
$\hat{y}(\text{Temp} = 15, \text{Insulation} = 5)$	357.7	403.22

Using the best-fit model from the cleaned data without outlier (Linear model):

$$\text{Oil} = 601.60 - 5.482 \cdot (15) - 23.232 \cdot (5) = 601.60 - 82.23 - 116.16 = 403.21$$

$$\boxed{\textbf{Predicted Oil Usage} \approx 403.2 \text{ gallons}}$$

```
Prediction at (15F,5in) using CLEANED Linear best-fit: 403.22 units
```

Figure 34: Simulation Results of Prediction Oil of Cleaned Data

The quadratic model can be expressed as:

$$\hat{y}_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i}^2 + \beta_4 x_{1i} x_{2i} + \beta_5 x_{2i}^2$$

Table 16: Quadratic model coefficients and prediction

	With Outlier (Best Fit)	**Without Outlier**
β_0	680	785.5
β_1	-6.17	-6.96
β_2	-41.72	-72.16
β_3	-0.012	-0.0034
β_4	0.24	0.2567
β_5	0.66	2.744
R^2	0.9741	0.9768
$\hat{y}(\text{Temp} = 15, \text{Insulation} = 5)$	410.52	407.35

Using the best-fit model from the original data with outlier (Quadratic model):

$$\text{Oil} = 680 + (-6.17)(15) + (-41.72)(5) + (-0.012)(15^2) + (0.24)(15)(5) + (0.66)(5^2) = 410.65$$

$$\boxed{\textbf{Predicted Oil Usage} \approx 410.65 \text{ gallons}}$$

```
Prediction at (15F,5in) using ORIGINAL Quadratic best-fit: 410.52 units
```

Figure 35: Simulation Results of Prediction Oil of Original Data